

(a) Controlled instrument

frozen recipe: SGD 0.1, cosine, 200 ep,
batch 128, crop+flip only

committed subset indices (per dataset,
per fraction) — version controlled

pinned filter bank (fingerprinted)



every intervention sees

byte-identical images

deviations quarantined, never headline

(b) MomentAux: the fused prior



fixed Gabor
energy bank

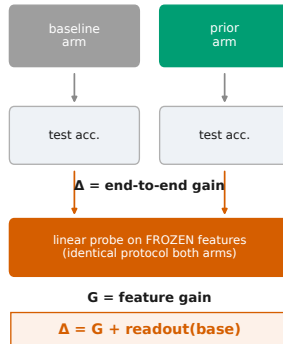
aux head
(train only)

$MSE \cdot \lambda$

+2% train compute · +0 at inference



(c) Two measurements per cell



(d) Comparators, by cost

- ImageNet transfer
external data
- SimCLR / SimSiam / DINO
2× compute
- FitNets learned teacher
2× compute
- DeiT augmentation
recipe change
- HOG target
1.02×
- MomentAux (ours)
1.02×

all fused into the SAME frozen recipe;
combinations measured too